COVER PAGE

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Competition Category: Advanced Process Technology Development Category (includes but is not limited to the following topics: Development of Next-Generation Memory Processes, Development of Logic Processes, Development of Third-generation Semiconductor Processes, Advanced Packaging Process Technology, Development of 3D IC Process Technology, High-Bandwidth Semiconductor Process Technology)

Competition topic: Rethinking Memory, Power, and Interconnect Design for Scalable AI Accelerators

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1. Project Title

3-D Stacked HBM and Compute Accelerators for Large-Scale AI: Co-Optimising Thermal Management, Interposer Materials, Geometry, and Power Delivery Efficiency

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2. Research Background & Motivation

The rapid emergence of large-scale generative artificial intelligence models, including large language models and multimodal systems, has significantly increased the demand for high-capacity, high-bandwidth memory and high-performance computation. Modern AI workloads for autonomous systems, cloud inference, and large-scale data processing require extremely high data throughput that traditional von Neumann architectures cannot efficiently support due to the physical separation between memory and compute units.

To overcome these limitations, the semiconductor industry has transitioned toward High Bandwidth Memory (HBM) and heterogeneous 2.5D/3D integration architectures. These systems place memory stacks in proximity to GPU or AI compute dies using interposers and through-silicon vias (TSVs), enabling terabit-per-second data transfer and significantly reduced latency.

However, this transition has shifted the primary performance bottleneck from data transfer latency to thermal management. In 3D integrated circuits, vertical stacking of heterogeneous components results in extremely high power densities. Current AI accelerators can exceed 700 W per device, with localised hotspots surpassing 500–1000 W/cm². These dense thermal regions are often trapped within stacked DRAM layers, leading to:

- Increased leakage current and reduced energy efficiency
- Thermal-induced DRAM refresh and retention degradation
- Thermo-mechanical stress and reliability concerns
- Reduced operating frequency due to thermal throttling

Studies indicate that thermal stress contributes to a significant proportion of electronic system failures in high-density packages. Without efficient heat removal pathways, temperature accumulation in memory stacks and interposer layers becomes a critical barrier to further scaling.

The interposer layer plays a central role in this architecture. Traditional silicon interposers provide electrical connectivity and structural support but exhibit limitations in lateral heat spreading under extreme AI workloads. Alternative materials such as glass offer improved signal integrity but suffer from extremely low thermal conductivity, while organic substrates provide limited thermal performance.

Hexagonal boron nitride (h-BN) and other anisotropic materials have emerged as promising candidates for next-generation interposers due to their exceptional in-plane thermal conductivity and electrical insulation properties. Combined with optimised TSV placement and geometry-based heat flow design, these materials may enable substantial reductions in hotspot temperature and power leakage.

This project aims to investigate thermal bottlenecks in 3D GPU–HBM systems and propose an integrated thermal-aware design methodology that improves reliability, energy efficiency, and scalability for next-generation AI hardware.

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3. Introduction

The integration of compute accelerators with stacked memory has become essential for meeting the computational demands of modern AI workloads. High Bandwidth Memory enables extremely wide I/O interfaces and short interconnect distances by vertically stacking DRAM dies connected through TSVs and mounted on interposers.

Despite these advantages, 3D stacking introduces complex thermal challenges. Heat generated in the AI accelerator must propagate through multiple memory layers, bonding interfaces, and interposer structures before reaching external cooling mechanisms. Due to limited lateral thermal conductivity within DRAM layers and interfacial thermal resistance between stacked components, heat tends to accumulate in central regions of the stack.

Thermal gradients within these systems can lead to:

- Uneven temperature distribution across memory layers
- Increased DRAM refresh rates due to temperature sensitivity
- Reduced retention time and increased leakage
- Power delivery inefficiencies due to temperature-dependent resistance
- Reliability degradation over long-term operation

The interposer is a critical component influencing both electrical and thermal performance. While silicon interposers are widely used due to fabrication compatibility, they exhibit limited lateral heat spreading capability relative to emerging anisotropic materials. Hexagonal boron nitride, for instance, demonstrates extremely high in-plane thermal conductivity while remaining electrically insulating, making it a promising candidate for advanced interposer design.

In addition to material selection, physical geometry plays a crucial role in thermal behaviour. Conventional flat interposers often experience central thermal trapping where heat accumulates beneath high-power compute dies. Geometry-based approaches such as variable-thickness interposers and edge-thinning can reduce thermal resistance toward package boundaries, enabling directional heat flow away from hotspots.

Furthermore, TSV placement strongly influences both electrical performance and heat conduction. TSVs serve as vertical thermal bridges between stacked layers, and non-uniform TSV distributions targeting hotspot regions may provide more efficient heat extraction than uniform layouts.

This project investigates the combined influence of interposer material properties, stack geometry, TSV distribution, and power delivery network co-design on thermal behaviour in 3D AI–HBM architectures.

4. Proposed Architecture: Hybrid Silicon + h-BN Interposer

To address the thermal and interconnect limitations identified above, this project proposes a hybrid 2.5D interposer architecture that augments a conventional silicon core with an integrated 100nm - 2μm CVD-grown hexagonal boron nitride (h-BN) functional layer. Rather than replacing the silicon interposer entirely — which remains infeasible at wafer scale given current CVD h-BN thickness limitations and grain boundary defect densities (TRL < 3) — the proposed design preserves the mechanical rigidity, TSV maturity, and CMOS compatibility of silicon while exploiting h-BN's uniquely dual-function material properties.

The h-BN layer serves three simultaneous roles: an ultra-low-k dielectric ($\varepsilon_r \approx 2.5$–$3$) for the redistribution layer, a high-breakdown vertical isolation layer (>7 MV/cm), and a lateral thermal spreading sheet. Its anisotropic thermal conductivity — approximately 400–600 W/mK in-plane versus only 2–5 W/mK out-of-plane enables heat to spread laterally beneath HBM stacks while minimising vertical thermal leakage between dies. Compared to the conventional SiO_2 interlayer dielectric ($\varepsilon_r = 3.9$, $k = 1.4$ W/mK), h-BN offers a roughly 31%

reduction in RDL parasitic capacitance and an improvement in lateral heat spreading resistance relative to silicon. At the system level, for a 1 kW AI accelerator package with 8 HBM stacks and aggregate interposer bandwidths exceeding 10 Tb/s, these gains translate to measurable watt-level energy savings and projected hotspot temperature reductions of 5–10°C margins that become increasingly critical as power densities scale super-linearly with model size.

h-BN occupies a rare position in the materials landscape as one of the only solids simultaneously offering low permittivity and high in-plane thermal conductivity. Alternative low-k dielectrics such as SiCOH or porous organics reduce capacitance but exhibit thermal conductivities below 0.5 W/mK, worsening hotspot accumulation. Conversely, thermally capable materials like AlN carry $\varepsilon_r \approx 8$–9, negating any signal integrity benefit. This dual-function advantage positions h-BN not as a simple dielectric substitute, but as a targeted co-optimisation layer for ultra-high bandwidth, high-power-density AI accelerator packages, precisely the class of system this project investigates.

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5. Experimental Design and Results

5.1 Objectives

The primary objective is to develop a simulation-driven framework to analyse and optimise thermal behaviour in 3D stacked AI accelerator–HBM systems.

Key research questions include:

1. How does heat accumulate and propagate in stacked GPU–HBM architectures?
2. Which parameters most strongly influence peak temperature and thermal gradients?
3. Can advanced interposer materials, such as hexagonal boron nitride, outperform conventional silicon?
4. How does TSV density and placement affect heat dissipation?
5. Can geometry-based interposer shaping reduce central thermal trapping?
6. How does power delivery network design interact with thermal behaviour?

5.2 Simulation Framework

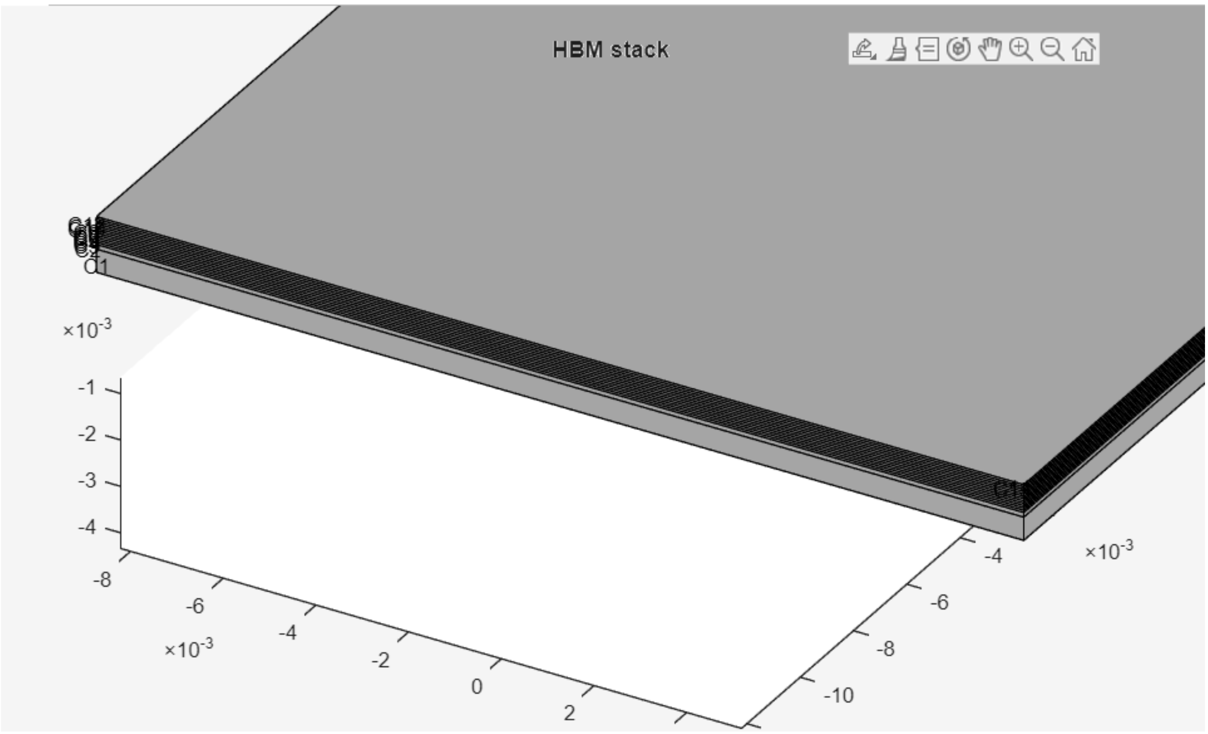
A three-dimensional steady-state thermal model is constructed to represent:

- AI accelerator compute die with a localised hotspot
- Silicon or advanced material interposer
- Multi-layer stacked DRAM (HBM)

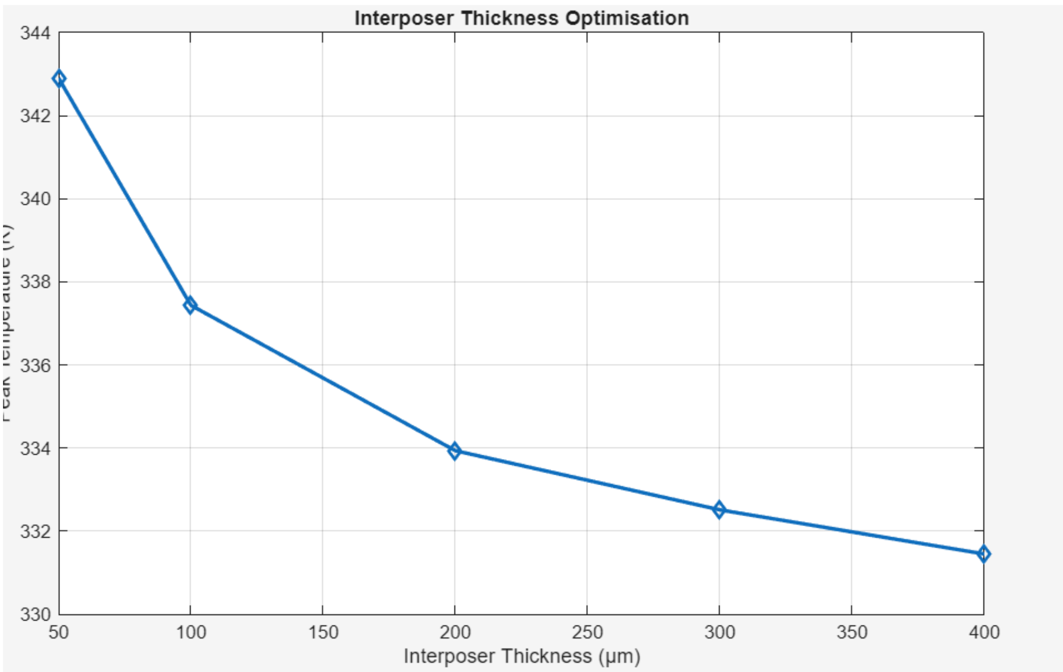
- TSV-based vertical heat paths
- External cooling boundary conditions

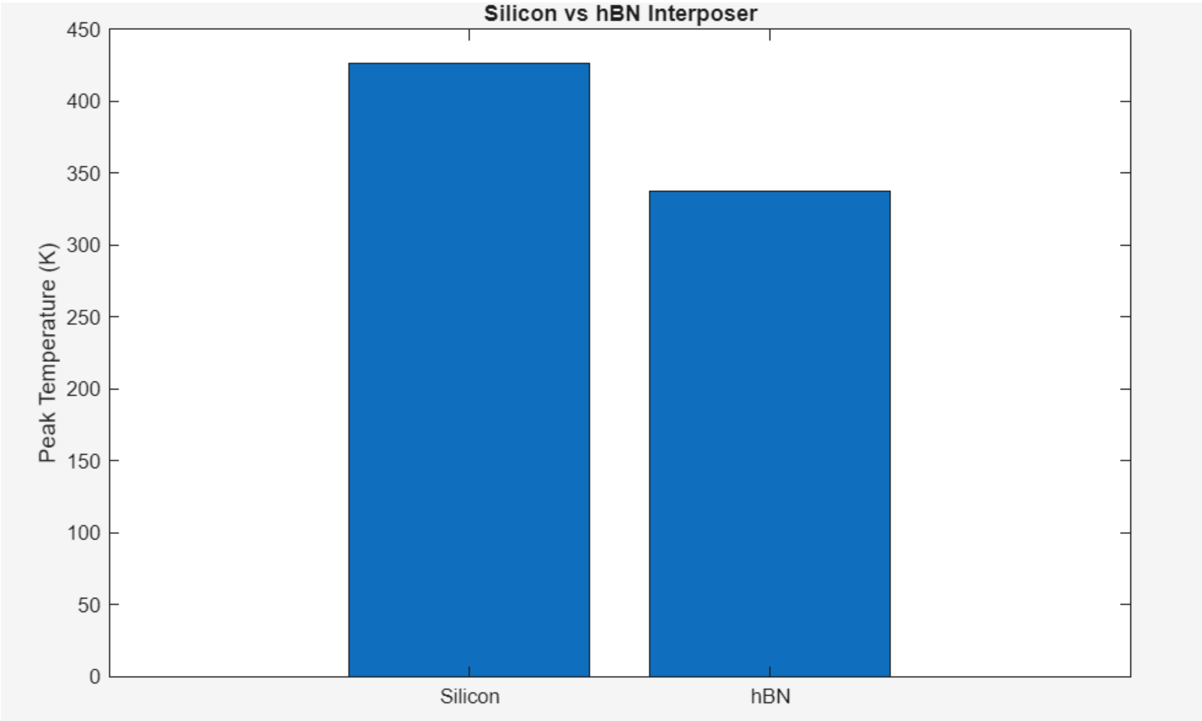
The model incorporates realistic material properties, anisotropic conduction behaviour, and localised heat generation to replicate practical operating conditions.

HBM in MATLAB:



MATLAB SIMULATION RESULTS

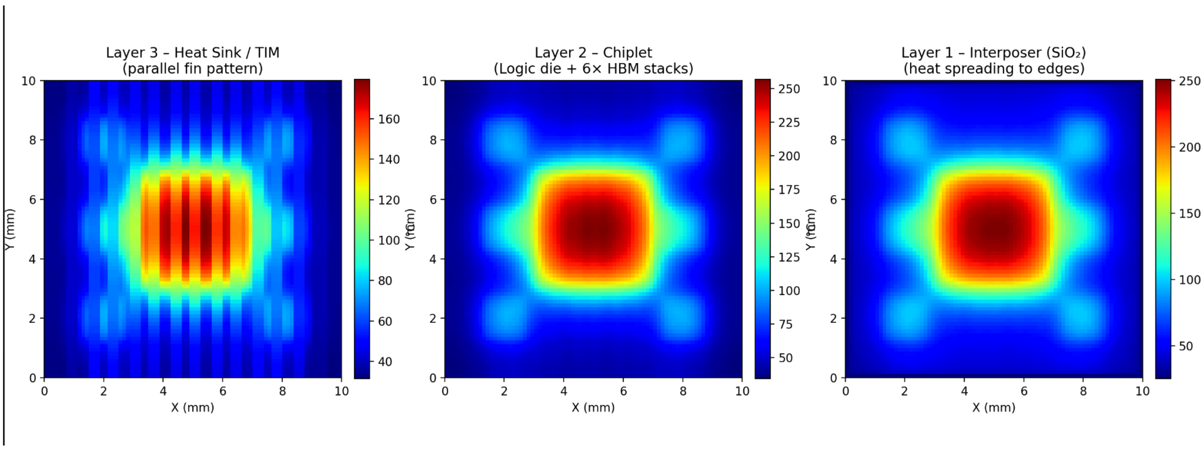




hBN gave a lower peak temperature as expected, as its in-plane conductivity is good.

TEMPERATURE DISTRIBUTION ON AI CHIPS:

Temperature distribution of a basic AI accelerator (logic-die + 6 HBM stacks) using Python.



Thermal Simulation: Uniform vs Hybrid Extraction Geometry

We implemented a 2D steady-state anisotropic thermal model to compare temperature distributions under different interposer geometry configurations. The governing equation solved was:

$$k_{\text{in}} \nabla^2 T - \frac{k_{\text{cross}}}{t(x,y)} T + Q = 0$$

where:

- T is temperature,
- k_{in} and k_{cross} are in-plane and cross-plane conductivities,
- $t(x,y)$ is local thickness,
- Q is the heat source.

The solver used a sparse linear system approach to compute the steady-state temperature field on a structured grid.

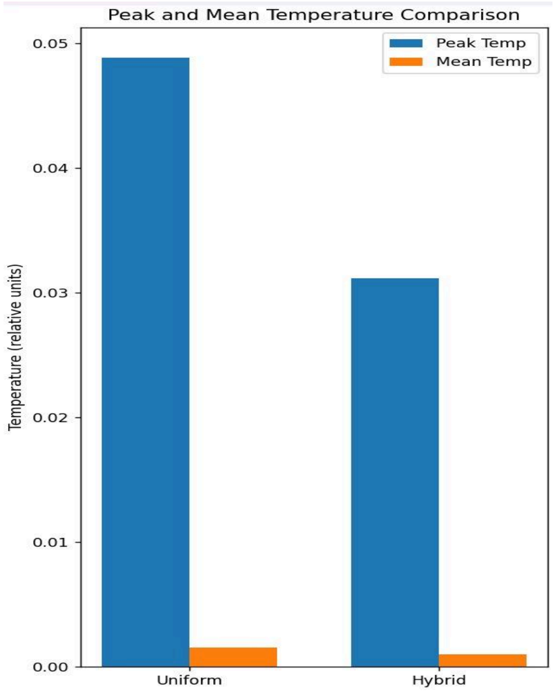
Cases Compared

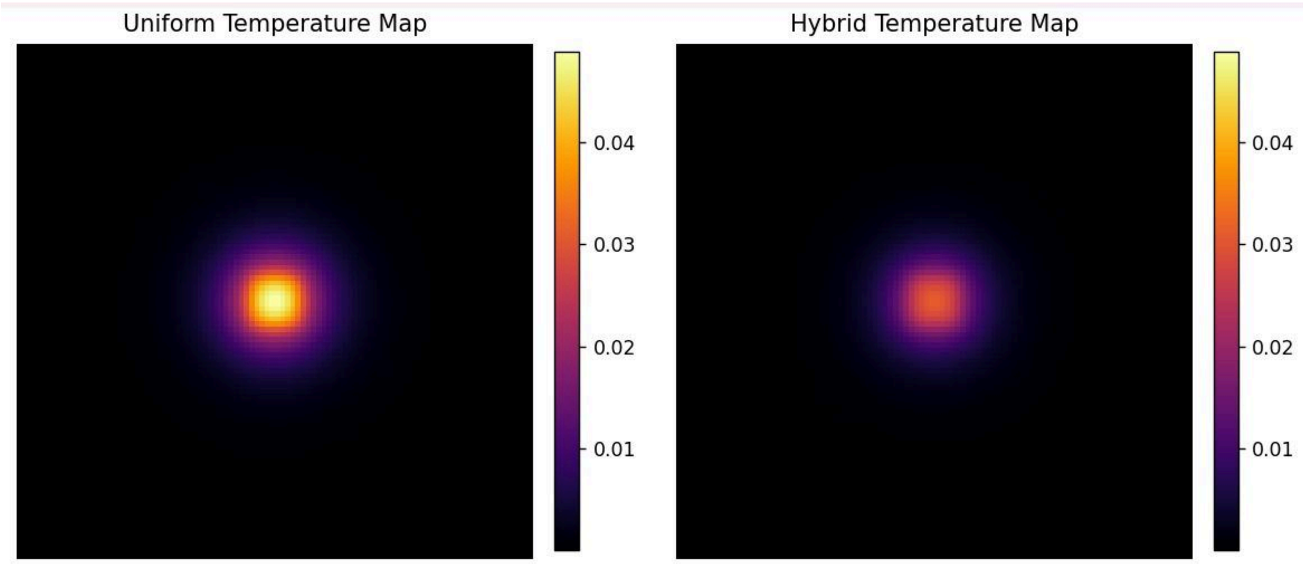
Two geometric cases were evaluated:

1. **Uniform Thickness (Baseline):**
   The interposer thickness was constant across the domain. Heat extraction relied only on uniform cross-plane leakage and lateral diffusion.

2. **Hybrid Extraction Geometry:**
   The thickness was reduced both under the central heat source and along the domain boundaries. This configuration enhances *vertical extraction* locally where heat generation is highest and at the edges, enabling heat that spreads laterally to escape more efficiently.

Local via density was kept zero to isolate geometric effects.





Key Results

For each case, the simulation computed the temperature field; key metrics extracted were:

- **Peak Temperature:** Maximum temperature in the domain
- **Mean Temperature:** Average temperature across the domain

The hybrid geometry produced a **lower peak and mean temperature** compared to the uniform case. This demonstrates that tailoring thickness to create focused vertical extraction regions, in combination with lateral spreading, can improve thermal performance.

Conclusion

The simulation study shows that a hybrid extraction geometry (combining local thinning under heat sources with thinner extraction zones at the boundaries) is more effective at lowering both peak and mean temperatures than a uniform interposer thickness. This suggests that purposeful geometric modulation can enhance thermal performance in anisotropic interposer materials.

5.3 Parameters Investigated

Material Parameters

- Silicon interposer thermal conductivity
- Hexagonal boron nitride anisotropic conductivity
- DRAM layer effective thermal properties
- TSV thermal conductivity

Geometrical Parameters

- Stack height and number of memory layers
- Interposer thickness and geometry shaping
- TSV placement (uniform vs centre-dense)
- Hotspot size and location

Thermal Boundary Conditions

- Top surface convection cooling
- Bottom substrate conduction
- Localised volumetric heat generation

Performance Metrics

- Peak temperature
- Temperature gradient across the stack
- Hotspot spread area
- Thermal resistance of the stack
- Relative improvement vs baseline silicon interposer

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6. Project Summary & Future Work

Project Summary

This project investigates thermal bottlenecks in next-generation AI accelerator–HBM systems and proposes a thermally co-optimised architecture integrating advanced interposer materials, geometry-aware heat spreading, TSV engineering, and power delivery network optimisation.

The proposed methodology treats thermal design as a primary architectural parameter rather than a secondary packaging constraint. By combining material innovation with geometry optimisation and electro-thermal co-design, the project aims to reduce hotspot temperatures, improve reliability, and enable higher compute density in future AI hardware.

Future Work

Future work will focus on validating and refining the proposed **Si interposer with a thin CVD h-BN functional dielectric layer** through improved simulation and practical feasibility studies.

1. **Advanced Thermal Simulation (Siemens EDA/Simcenter):**
   We will implement the current analytical and FEM models into Siemens tools to perform higher-fidelity 3D electro-thermal simulations of the TSV–hBN–RDL stack. The goal is to compare temperature rise, heat spreading, and hotspot formation between SiO_2 and h-BN under realistic HBM power densities.

2. **Geometry Sensitivity Studies:**
   Conduct controlled parametric studies by varying only key parameters such as h-BN thickness, Layer Geometry, TSV pitch, and power density. The objective is not redesign, but quantitative comparison to better understand where thermal improvements are meaningful.

3. **CVD h-BN Feasibility (College Facilities):**
   Attempt small-scale CVD growth of h-BN films to evaluate thickness uniformity, surface quality, and basic dielectric behavior. This will help assess practical integration challenges.

4. **Refined Reliability Modeling:**
   Improve electro-thermal coupling models by incorporating experimentally reported breakdown fields and leakage behavior to evaluate safety margins under high-bandwidth operation.

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7. References

1. Thermal Analysis of 3D GPU-Memory Architectures with Boron Nitride Interposer
2. JEDEC High Bandwidth Memory (HBM) Standards
3. Thermal Issues Related to Hybrid Bonding of 3D-Stacked High-Density Chips
4. IEEE Transactions on Components, Packaging and Manufacturing Technology
5. Research on anisotropic thermal conduction in hexagonal boron nitride
6. Recent studies on electro-thermal co-design in AI accelerator packages

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